Shenzhen Hiwonder Technology Co,Ltd

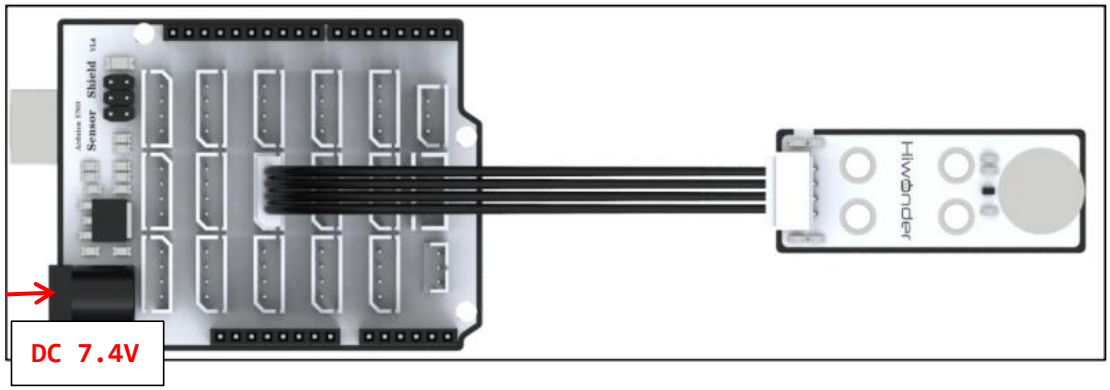
Arduino Development Tutorial

V1.0

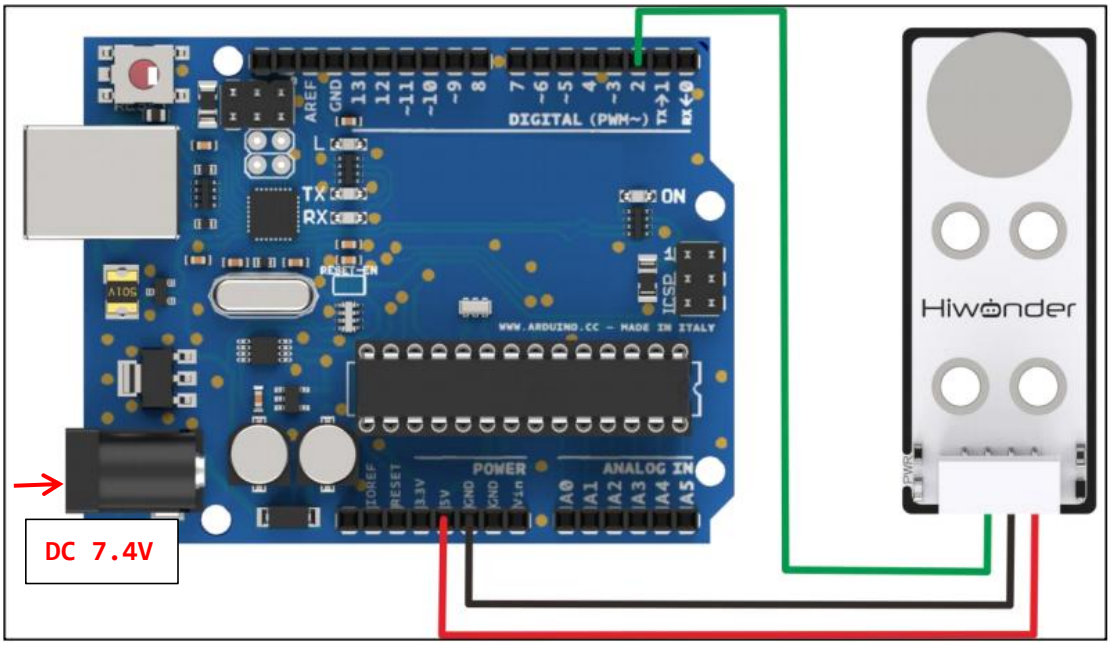
1. Getting Ready

1.1 Wiring Instruction

This section illustrates connecting to pins A2 and A3 on the Arduino expansion board using a 4-pin cable. Please refer to the diagram below:



If you don't have Arduino expansion board, please directly connect the sensor to the Arduino demo board with DuPont wire, as shown in the following connection diagram:



Note:

① If you are using the lithium battery provided by our company, please connect its red wire to the positive terminal (+) of the DC port, and its black wire to the negative

terminal (-).

② If the lithium battery wires are not connected, do not directly connect them to other wires to avoid a short circuit between the positive and negative poles, which may cause damage to the device.

③ Before powering on, please ensure no metal objects come into contact with the master. This action prevents the pins at the bottom of the master from a short circuit, which may cause damage to the master.

1.2 Environment Configuration

To install Arduino IDE on your PC, you can access the software package in “2. Software Tools&Program Collections->01 Software Installation Packages-> Arduino Installation Package”. For detailed instructions on how to use Arduino IDE, please refer to the corresponding directory.

2. Case Test

This case uses a terminal window to display the detecting status of the touch sensor.

2.1 Program Download

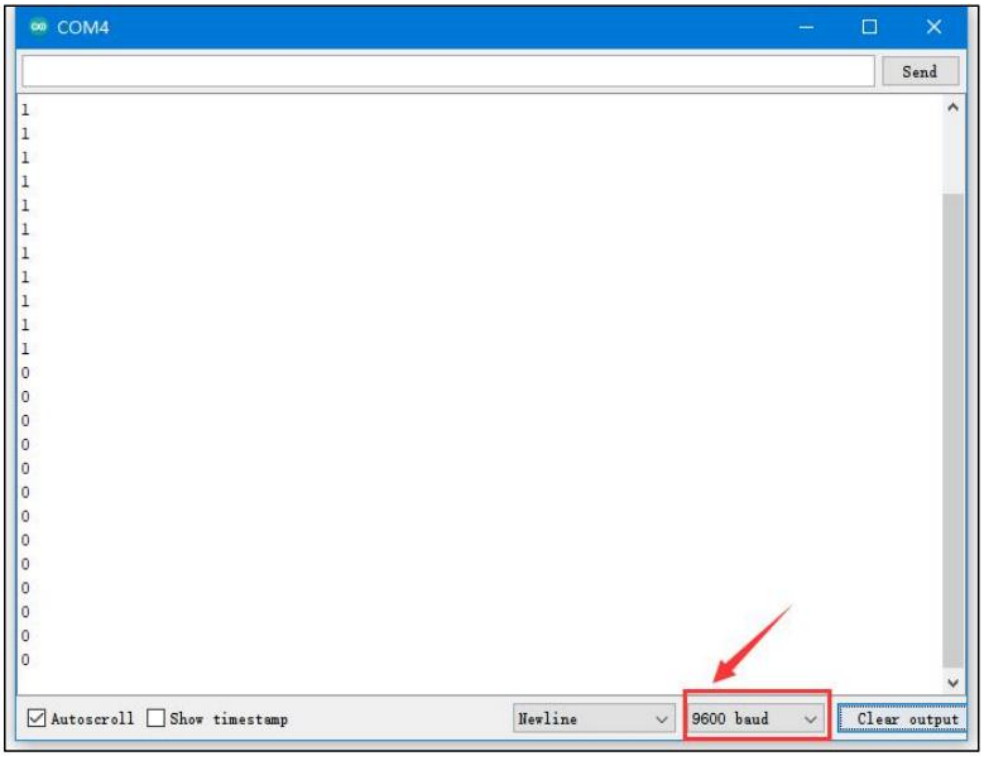
1) Connect the Arduino UNO demo board with the expansion board to the computer using a USB cable. Open Arduino IDE and click “File->New” to import the program from this tutorial directory.

2) Select the suitable demo board and port (the following port is for reference only, please rely on the actual connection), then compile and upload the program.

3) After the code is uploaded successfully, click to open the serial monitor and set the baud rate to 9600. Then, observe the phenomenon.

2.2 Program Outcome

Use the touch sensor to test the touch state. When you do not touch the metal surface and the sensor does not detect any touch, the serial monitor displays “1”. When you touch the metal surface with your finger, it detects the touch and its LED1 lights up, and the serial monitor displays “0”.



2.3 Brief Program Analysis

- **Port Definition**

```
    #define TOUCH 2    //Define the digital port 2 connecting to the signal port of the touch sensor

    uint8_t touch_value;
```

Initially, the interface of the touch sensor is defined, with the OUT port connected to the D2 pin. The “touch_value” stores the output value of the touch sensor.

- **Serial Initialization**

```
    void setup()

    {

      Serial.begin(9600);

    //Configure Touch as input (input status is to read the pin, that is, to read the feedback value of the sensor

      pinMode(TOUCH, INPUT);
```

```
}
```

After the serial initialization, set the D2 pin to input mode.

- **Loop Execution**

```
void loop()
{
  touch_value = analogRead(TOUCH);    //Collect the output value of the touch sensor
  Serial.println(touch_value);
  delay(500);
}
```

Collect and output data in the “loop()” function.

Use “digitalRead” to read and output the D2 level signal to the “touch_valye” variable to obtain the high-low level signals. Then, the sampling outcomes are printed on the serial monitor. Please touch the metal surface repeatedly to observe the changes in the data printed on the serial monitor.